

VanRakshak AI

1. Project Overview

VanRakshak AI is an autonomous wildlife intelligence platform built for forest surveillance, anti-poaching, illegal logging detection, encroachment monitoring, and early fire response. The system combines autonomous drone patrols, multi-sensor perception, edge AI inference, and intelligent alerting so forest authorities get actionable intelligence instead of raw video.

- Primary use case: forest protection and wildlife conservation.
- Core capability: onboard AI-based threat detection and real-time alerts.
- System design goal: low-latency, low-bandwidth, field-ready operation in remote terrain.
- Roadmap: prototype -> pilot -> multi-division deployment -> broader critical-infrastructure use.
- Regulatory pathway: designed for line-of-sight, geofenced patrol operation compliant with DGCA's current drone rules, with BVLOS (beyond visual line of sight) clearance planned as a Phase 2 requirement.

2. Overall System Architecture

This diagram shows the end-to-end flow from mission planning to field response. The drone collects data from its sensor suite, edge AI processes the inputs, and alerts reach the forest command dashboard and ranger team.

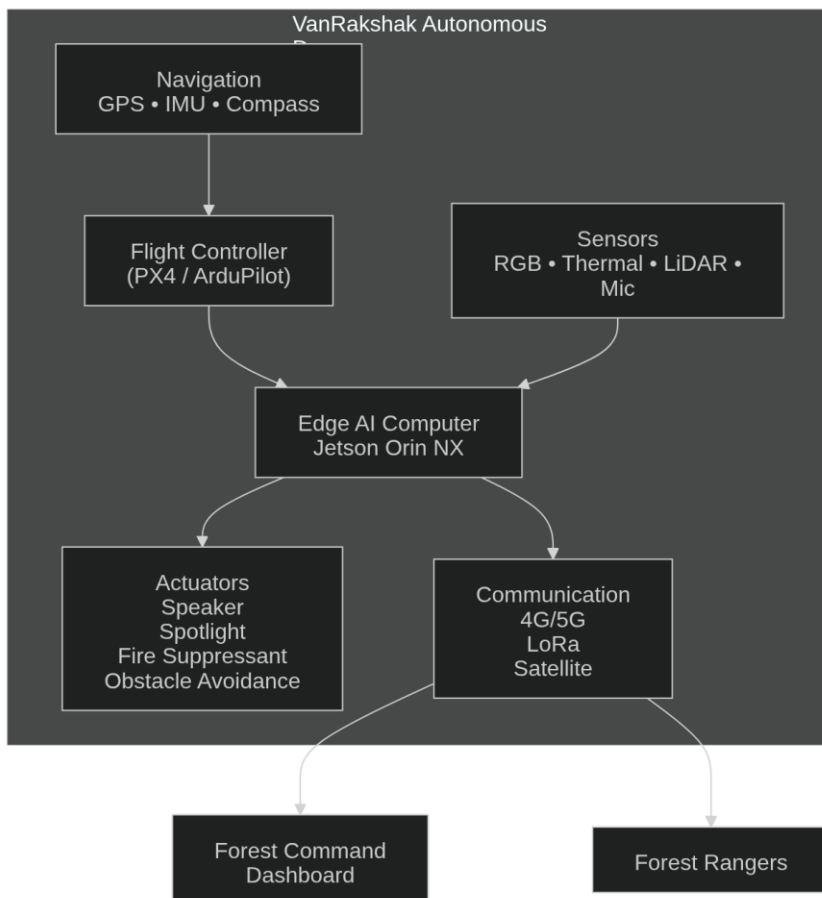


Figure 1. High-level VanRakshak AI system architecture.

The architecture is intentionally modular. Navigation, sensing, inference, communication, and response can be improved independently without changing the entire platform.

3. Sensor Fusion and AI Perception

The perception stack fuses RGB, thermal, LiDAR, and acoustic data. Detection models identify humans, animals, vehicles, smoke, fire, and other suspicious events. Tracking and classification keep the system aware of what is happening over time, not just in a single frame.



Figure 2. Sensor fusion and AI perception pipeline.

The perception layer is the core intelligence layer. It reduces false alarms by combining multiple signals and by tracking objects across time before generating an alert.

4. Agentic Decision Pipeline

This pipeline shows how detected events are contextualized using mission memory, scene understanding, safety rules, and forest policies before the system selects a response. The aim is not just detection, but decision support.



Figure 3. Agentic decision pipeline from detection to action selection.

In prototype form, this layer should stay conservative: classify, score, alert, and track. More advanced planning and policy-based actions can be added in later versions.

5. Autonomous Mission Loop

This loop shows the patrol cycle running onboard during a mission. The drone scans the environment, detects objects, analyzes the situation, computes risk, and then chooses the appropriate response.

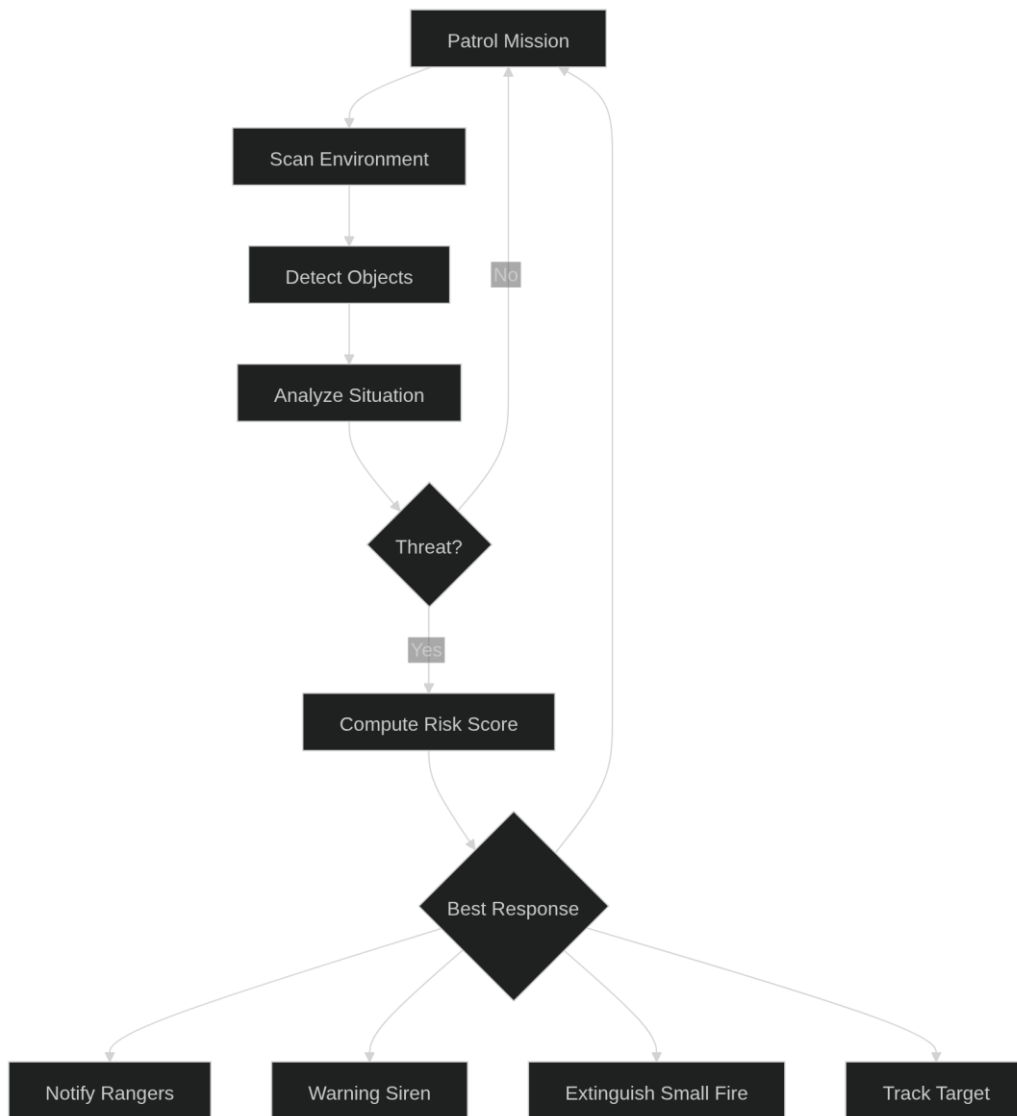


Figure 4. Autonomous decision loop for repeated patrol cycles.

For a submission-grade prototype, the safest high-value outputs are alerting rangers, tracking targets, and supporting fire warnings. Fire suppression can be presented as a future or optional payload, not a baseline claim.

6. Multi-Drone Collaboration

This slide extends the concept from a single drone to a coordinated drone fleet. One drone detects smoke or intrusion, another verifies the event visually, and a third can act as a communication relay in low-connectivity areas.



Figure 5. Cooperative multi-drone operating model.

This is valuable for larger protected areas because it improves coverage, redundancy, and alert confidence.

7. Hardware Reference View

The detailed drone blueprint shows a realistic payload-first approach: camera suite, compute module, siren, spotlight, and modular bay. This is a useful reference for the prototype build.

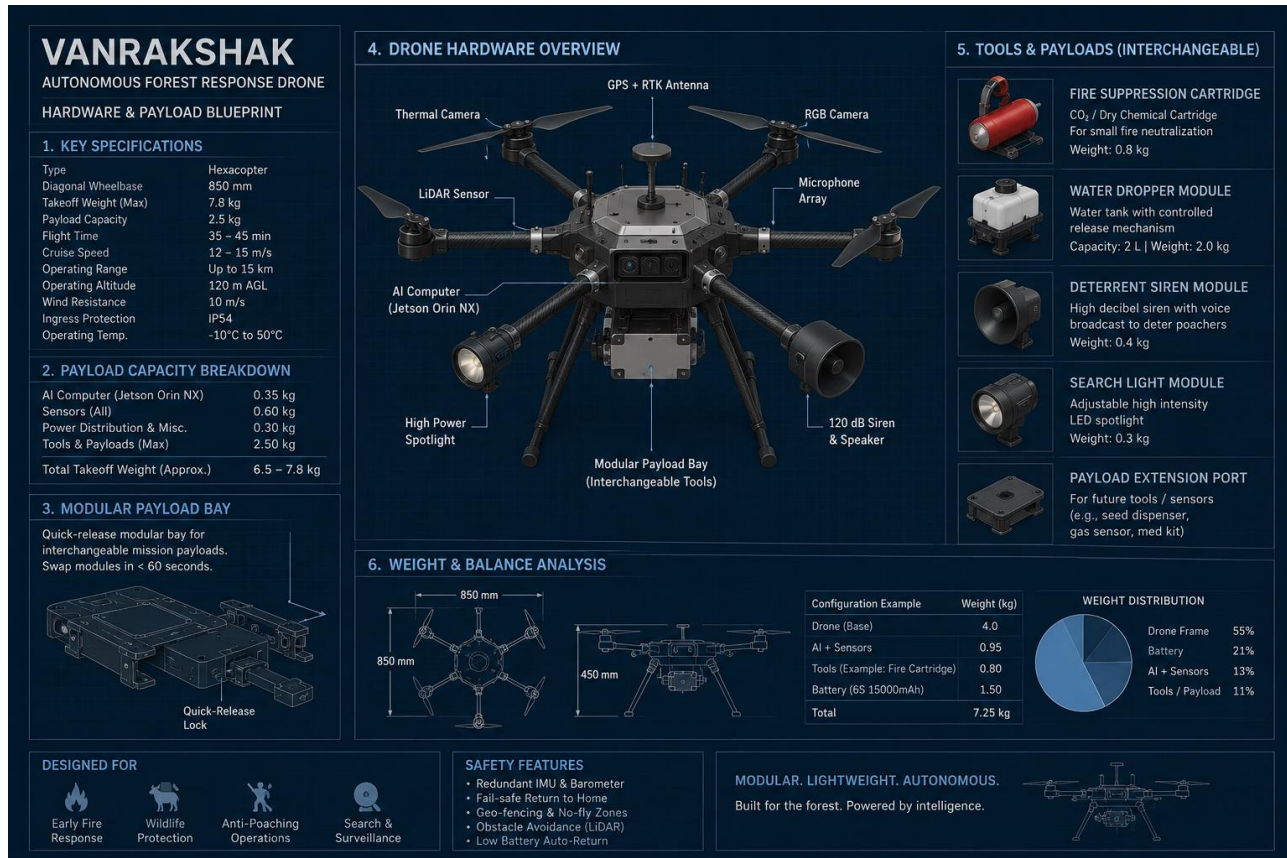


Figure 6. Hardware blueprint and interchangeable payload concept.

For the near-term prototype, the most defensible stack is an airframe, flight controller, Jetson-class compute, RGB camera, thermal camera, LiDAR, GPS/IMU, and a basic alert payload.

8. Commercial Pathway and Manufacturing Potential

VanRakshak AI can be positioned as an indigenous protection-tech platform with a clear buyer, a serviceable pricing model, and a phased manufacturing roadmap.

Target Buyer

- State Forest Departments for protected areas, tiger reserves, wildlife sanctuaries, and reserve forests.
- Secondary buyers: private wildlife sanctuaries and conservation NGOs.
- Procurement route: state forest budgets plus centrally sponsored schemes such as Project Tiger, Project Elephant, and CAMPA-linked forest protection budgets.

Business Model

- Drones-as-a-Service: departments pay for patrol coverage per hectare or per month instead of buying all hardware upfront.
- The model reduces the capital burden on public agencies and shifts the spend toward operational expenditure.
- Scale path: pilot with one forest division -> validate performance -> expand to multi-division contract -> state-wide rollout.

Cost and Unit Economics

- Estimated unit cost: ₹2.5-3.5 lakh per drone for airframe, compute, sensors, and payload modules.
- Estimated service pricing: ₹15,000-25,000 per sq. km per month of active patrol coverage.
- Value proposition: one drone can cover significantly more area per cycle than manual foot patrols at lower recurring headcount cost.

Manufacturing and Import Substitution

- Phase 1: source compute modules and premium sensors from existing suppliers to get the prototype moving.
- Phase 2: localize the airframe, payload bay, siren/speaker module, and power electronics with domestic manufacturing partners.
- Target: 60%+ domestic value-add in Phase 2, aligned with Make in India sourcing goals.

Export Potential

The concept is not India-only. Similar poaching, logging, and forest-fire challenges exist in Southeast Asia and Africa, where ranger coverage is often under-resourced. That gives VanRakshak AI a credible export path if the prototype proves itself.